Prabhavit Gupta

Solution Architect | LangGraph, Agentic AI, Multi-Agent Systems & Cloud Integrations

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Executive Summary

- **LangGraph & Multi-Agent Solution Architecture:** AI Solution Architect with 3 years commercial software experience specializing in architecting, developing, and governing stateful LangGraph workflows, multi-agent orchestrations, and autonomous system integrations across enterprise environments.
- **Deterministic Boundaries & Human-in-the-Loop Governance:** Expert in decomposing ambiguous business processes into robust cyclical graph topologies, checkpointed state persistence, deterministic validation gates, human approval escalation flows, and secure tool access via Model Context Protocol (MCP).
- **Production Cloud Engineering & Observability:** Proven track record deploying scalable agentic backends on GCP (Cloud Run, GKE, Vertex AI) and AWS, establishing full-stack telemetry and evaluation harnesses with LangSmith, OpenTelemetry, and DeepEval to guarantee enterprise reliability, latency SLAs, and zero data leakage.

Technical Skills & Core Competencies

- **Agentic Frameworks & Architecture:** LangGraph, LangChain, Multi-Agent Coordination (Supervisor, Hierarchical, Peer graphs), StateGraph, State Persistence (Postgres/Redis Checkpointers), Cyclic Workflows, Human-in-the-Loop Interrupts.
- **Enterprise Integration & Tool Calling:** Model Context Protocol (MCP), OpenAPI / REST Tool Schema Synthesis, Function Calling, Pydantic Structured Outputs, OAuth2/OIDC, Secrets Management, Event-Driven Webhooks.
- **LLM Models & Inference Orchestration:** Anthropic Claude (Sonnet/Opus), OpenAI (GPT-4o), Google Gemini, DeepSeek, Local Hugging Face / vLLM, Context Window Pruning, Dynamic Prompt Caching.
- **RAG & Vector Architecture:** Hybrid Retrieval (Dense Vector + BM25 Sparse), Pinecone, Qdrant, ChromaDB, Sentence-Transformers, Chunking Strategies, Metadata Filtering, Context Re-ranking (Cohere).
- **Cloud, DevOps & Infrastructure:** Google Cloud Platform (Cloud Run, Cloud Functions, Pub/Sub, Vertex AI), AWS (ECS, Lambda, S3), Docker, Terraform, GitHub Actions CI/CD, Asynchronous Task Queues (Celery/Redis).
- **Security, Governance & Observability:** LangSmith, OpenTelemetry, Prometheus, DeepEval, RAGAS, Prompt Injection Defense, PII Masking, Role-Based Access Control (RBAC), Audit Logging.

Professional Experience

AI Solutions Architect & Lead Engineer • *Independent / Self-Employed Consultancy*
Oct 2024 – Present • Remote / Venice, Italy

- Architected and deployed multi-agent enterprise automation solutions using LangGraph, integrating CRM, ticketing, and analytical databases with deterministic state graphs, reducing manual processing overhead by 70%.
- Designed resilient state persistence and recovery mechanisms using PostgreSQL checkpointing, allowing long-running multi-turn agent processes to pause for human approvals and resume seamlessly without token loss.
- Engineered standard Model Context Protocol (MCP) servers and clients, establishing secure sandboxed interfaces for agent tool invocation against internal REST APIs while enforcing strict schema validation and parameter sanitization.
- Formulated architectural blueprints separating non-deterministic LLM reasoning from critical transactional operations via deterministic validation barriers, preventing hallucinated API mutations.
- Implemented continuous evaluation pipelines using LangSmith and custom synthetic benchmark suites, tracking agent task completion rates, cost per run, and execution latency across model updates.

AI Software Engineer • *Freelance & Enterprise Client Projects*
Jan 2023 – Sep 2024 • Hybrid / Italy

- Designed and shipped production RAG platforms integrating multi-stage document ingestion, semantic chunking, and hybrid search (vector + BM25) over unstructured enterprise knowledge repositories.
- Integrated LLM reasoning agents into asynchronous backend architectures utilizing FastAPI, Celery, Redis, and GCP Cloud Run, supporting concurrent client sessions with sub-second response times.
- Built automated regression testing harnesses using DeepEval and golden evaluation datasets, catching prompt regressions and edge-case failures prior to production deployment.
- Implemented robust guardrails and PII redaction layers adhering to EU GDPR and enterprise data governance standards, mitigating prompt injection risks and accidental data exposure.
- Collaborated with business stakeholders to define AI functional specifications, translating complex workflow requirements into modular microservices and maintainable system designs.

Featured Architecture & System Implementations

Enterprise Multi-Agent Orchestrator (LangGraph & MCP) • *Lead Architect*
Open-Source / Production Reference • 2025 – 2026

- Designed a modular multi-agent system utilizing LangGraph StateGraph, featuring a supervisor agent dynamically routing requests between specialized researcher, code generation, and audit sub-agents.
- Integrated state checkpointing with Postgres and human-in-the-loop interruption gates for sensitive administrative actions, enabling operational transparency and complete auditability.
- Deployed the entire orchestration cluster on GCP Cloud Run with LangSmith distributed tracing, achieving 99.4% execution reliability across multi-step planning tasks.

High-Throughput Enterprise RAG with Semantic Cache • *Architecture & Build*
Production Reference • 2024

- Architected a high-concurrency enterprise retrieval system utilizing Qdrant vector store and Redis semantic caching, decreasing API inference costs by 38% and response latency by 45%.
- Built metadata filtering pipelines and reciprocal rank fusion (RRF) re-ranking, boosting retrieval precision on complex financial and technical documentation.

Education, Credentials & Languages

- **Bachelor of Arts in Philosophy & Economics** • Ca' Foscari University of Venice (Focus: Formal Logic, Decision Theory & Analytical Systems)
- **Languages:** English (C1 – Full Professional Proficiency), Italian (Working Proficiency), Hindi (Native)
- **Work Authorization:** EU Permanent Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) • Partita IVA ready for freelance/contract engagement.